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Social and Environmental Factors Influencing Mother-Infant Separation-Reunion in Squirrel Monkeys

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JORDAN, T. C., M. B. HENNESSY, C. A. GONZALEZ AND S. LEVINE. *Social and environmental factors influencing mother-infant separation-reunion in squirrel monkeys.* PHYSIOL BEHAV 34(4) 489–493, 1985.—Sixteen squirrel monkey mother and infant dyads were housed in groups of two. They were separated and then immediately reunited in their familiar home cage or in a novel environment, either in conjunction with the second mother-infant dyad, or as a single manipulated pair. Behavioral observations of both dyads were made during the 30-min period following these manipulations; blood samples were then taken for cortisol assay. Mothers' locomotor activity increased when a single dyad was manipulated in the home cage and decreased in the novel environment where proximity between the pairs increased. Cortisol levels varied significantly across conditions, with mothers and infants responding to different stimuli. Mothers responded primarily to social disruption, including separation from the other dyad, and infants responded most clearly to novelty and separation from the other dyad. No significant differences were observed over base levels when both mother and infant pairs were reunited in the home cage, indicating that changes were not due to the disturbance involved in the separation procedure.

Mother-infant Separation-reunion Squirrel monkeys Plasma cortisol

PRIMATE infants are particularly sensitive to involuntary separation from their mothers, and typically, behavioral changes in response to disruption are used in assessing the impact of separation [1, 3, 8, 9, 10]. There are also various physiological consequences of mother-infant separation [14]. Activation of the pituitary-adrenal system has been used as one physiological indicator of the effects of brief separation in both mothers and infants. Such measures may reflect aspects of the state of the animal not readily apparent in its behavior [7]. Mothers and infants separated from each other for 30 min display plasma levels of cortisol that are dramatically elevated over resting levels [12]. However, cortisol levels observed 30 min following separation and immediate reunion with the mother are not elevated over basal values [13]. This separation-reunion phenomenon is of considerable interest, particularly when the maternal response is compared with that of nonlactating females in which significant increases of plasma cortisol concentrations were observed 30 min after handling [13]. It appears that the presence of the other member of the mother-infant pair may have inhibited the pituitary-adrenal response to capture and the disturbance of the separation procedure. Moreover, this would imply that some portion of the difference in cortisol levels observed in the complete separation and separation followed by immediate reunion conditions resulted from the absence of this social inhibition in the complete separation condition.

This apparent capacity of the mother and infant to inhibit cortisol responses raises several questions related to the generality of this phenomenon. One concerns the range of situations in which inhibition of pituitary-adrenal activity in mothers and infants can be affected by their mutual contact. Since novelty has been shown to be a potent elicitor of pituitary-adrenal activity, would, for instance, mother-infant contact inhibit the cortisol response to disturbance combined with exposure to a novel environment? Further, because squirrel monkeys live in complex social organizations [2], another concern is how other social partners affect the cortisol response. For example, if a mother and infant were to live in close proximity with another mother-infant pair, would the presence of this second pair serve to inhibit pituitary-adrenal responses in the first pair during mutual disruption, or would social disruption of one dyad augment the behavioral and physiological responses of the other pair?

The present study was designed to examine these issues and their effect on plasma cortisol and behavior. Mother-infant pairs were housed in groups of 2 in small primate cages for several weeks and various experimental conditions were then imposed. These conditions involved separating, and then immediately reuniting, mother-infant pairs in their familiar home cage, or in a novel environment—either in conjunction with the second mother-infant pair, or as a single manipulated dyad.

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METHOD

Animals and Housing

The animals used in this study were 16 pairs of squirrel monkey (*Saimiri sciureus*) mothers and infants of the Guyanese variety. The infants were born and reared in laboratory social groups until 1 month prior to the experiment (i.e., at 2 months of age), when pairs of familiar females with their infants were housed in standard individual primate cages (60×66×74 cm). Half of the cages were located in a single rack at one end of a long building and the other half in a second rack at the opposite end of the building. The cages faced toward a screen with windows covered with single vision film for observation purposes.

Procedure

Each subject was tested once in each of the conditions listed below. The order of condition presentation was counterbalanced across animals. The conditions were: (0) Base, in which the subjects were undisturbed and were observed for 30 min when blood was then sampled; (1) Separation and immediate reunion of both mother-infant pairs in their home cage; (2) Separation of both mother-infant pairs followed by immediate reunion in a similar cage in an adjoining room (the room was unfamiliar to the subjects and contained no other animals); (3) Separation-reunion of a single mother-infant pair in the home cage (leaving the other pair unmanipulated); (4) Separation-reunion of a single mother-infant pair in the unfamiliar environment. The remaining unmanipulated pairs of mothers and infants from the latter two conditions were also observed and blood samples were taken, constituting Conditions 3A and 4A, respectively.

The procedure for the separation-reunion manipulations involved hand capture of the mother and infant, briefly holding them apart, and then reuniting them either in the familiar or novel environment. Behavioral observations were made of both mother-infant pairs during the 30 min following this manipulation, and blood samples were then obtained from all 4 animals. Conditions were imposed at 1-week intervals, and no more than one cage of mother-infant pairs from each rack were tested on any day. All testing was conducted between 0900 and 1100 hr.

The categories of behavior scored were: Mother-Infant Contact—dorsal or ventral; Proximity Between Dyads (≤10 cm); Infant- or Maternal-Initiated Break in Contact; Retrieval or Reunion; Vocalization; and Movement. The first four categories are measures of interactions between animals, whereas Vocalization and Movement were included as behavioral signs of agitation or distress. Mother-Infant Contact and Proximity Between Dyads were noted every 30 sec as signaled by a click from an electronic timer; the frequencies of other behaviors were recorded throughout the 30-sec intervals. A Movement was scored each time the animal being observed crossed the vertical mid-line of the cage. Each mother-infant pair was scored by a separate observer. Observer ratings were tested for reliability, and a correlation coefficient of .97 was obtained.

The blood samples (0.5 ml) were collected in heparinized syringes under ethyl ether anesthesia by cardiac puncture. Two teams of two or three individuals each were used to catch the animals and collect blood samples rapidly. Using this procedure, the blood samples from all 4 monkeys tested in each session were collected within 3 min of the entry of the blood sample teams into the animal area. These precautions were employed so that all blood samples could be collected before the capture and blood-sampling procedure induced any appreciable rise in plasma cortisol levels [6]. The plasma was extracted, then frozen for subsequent cortisol radioimmunoassay. The cortisol assay method used was that previously described by Klemm and Gupta [11].

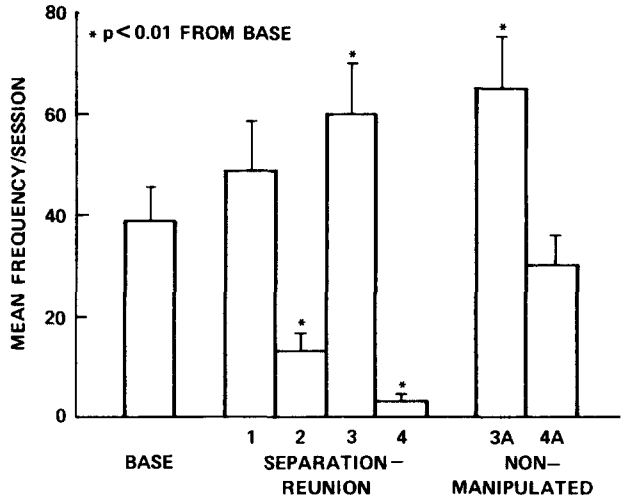


FIG. 1. Mean frequency (±SEM) of maternal movement during 30-min period following separation-reunion procedures (N=16).

Behavioral and cortisol measures were tested in separate 1-way analyses of variance (ANOVAs) for the mothers and infants. Post hoc tests were made using the Newman-Keuls test.

RESULTS

Behavior

The infants were in almost constant dorsal contact with their mothers under all conditions and vocalizations were infrequent. Consequently, the behavior categories of Infant- or Maternal-Initiated Break in Contact, Retrieval or Reunion, Vocalization by infants and mothers, and Movement by infants occurred too rarely for statistical analysis. Mother-Infant Contact occurred at near maximal levels in all conditions, so it too was not analyzed.

Maternal movement showed a significant effect across conditions, $F(6,90)=11.21$, $p<0.001$, as illustrated in Fig. 1. Post hoc analysis indicated that for Condition 1, in which both mother and infant pairs were separated and then immediately reunited in their home cage, levels of movement did not differ from those in the Base (undisturbed) condition. However, when a single pair was manipulated in the familiar environment, there were significant increases in the activity of the manipulated mother (Condition 3, $p<0.01$), as well as her nonmanipulated partner (Condition 3A, $p<0.01$). In contrast, when the mothers were manipulated and placed in a novel environment, movement virtually ceased (Conditions 2 and 4, $p<0.01$). The nonmanipulated female left alone with her infant in the familiar environment did not show any significant change in activity level (Condition 4A).

A significant difference was found for proximity between pairs of mothers and infants, $F(4,28)=10.10$, $p<0.001$. This was attributable to an increase in proximity in the novel environment (Condition 2) as compared to the base and all

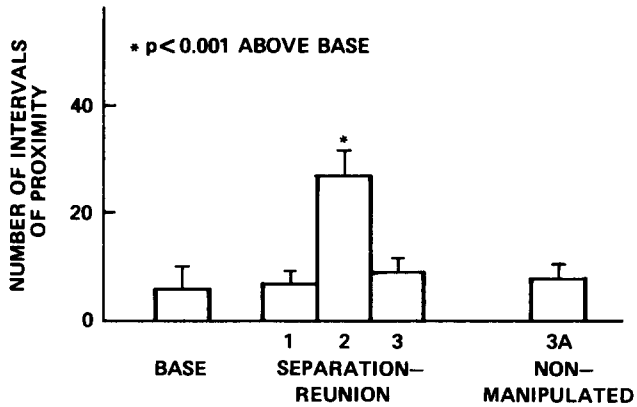


FIG. 2. Mean number (±SEM) of periods of proximity between mother-infant pairs following separation-reunion (N=8 pairs of dyads).

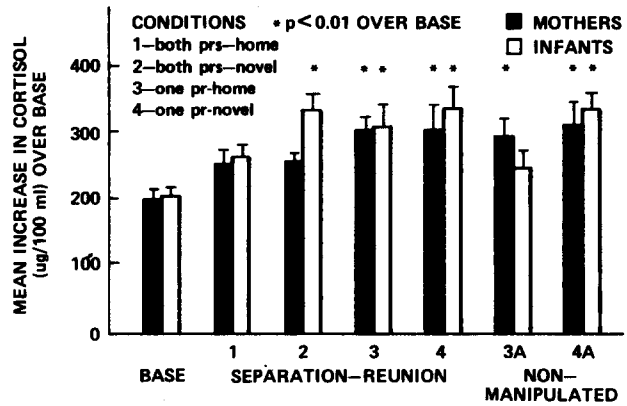


FIG. 3. Mean increase (±SEM) in cortisol (μg/100 ml) over base level for 16 mothers and infants following separation-reunion.

other conditions in which both pairs were observed in the same cage (see Fig. 2).

Cortisol

The plasma cortisol values for mothers differed significantly across conditions, $F(6,90)=4.36$, $p<0.001$. Subsequent analysis showed that when both mothers with their infants were manipulated and returned to the home environment, no significant increase over base was observed (Condition 1, Fig. 3). However, when only one mother-infant pair was manipulated, a significant increase over base was seen (Condition 3, $p<0.01$), and so, too, was a significant elevation observed in the corresponding nonmanipulated female partner (Condition 3A, $p<0.01$). The nonmanipulated mother also exhibited a significant increase in cortisol when her adult partner was removed to the unfamiliar room (Condition 4A, $p<0.01$). When both pairs of mothers were separated and immediately reunited with their infants in the novel environment, no increase in cortisol levels was seen (Condition 2). A significant elevation was, however, observed when only one pair was placed in the novel environment (Condition 4).

Plasma cortisol levels of infants also differed significantly across conditions, $F(6,90)=3.83$, $p<0.01$. Infants, like mothers, showed no increase in cortisol levels when both pairs were separated from their mothers and then immediately reunited in the familiar home cage (Condition 1). However, in contrast to their mothers, infants did respond with significant elevations when both pairs were placed in the novel environment (Condition 2, $p<0.01$). Infant cortisol levels were also increased when the infant and its mother, but not the other pair, were manipulated and placed in the novel environment (Condition 4, $p<0.01$). The infants also responded when they and their mothers only were separated and reunited in the home cage (Condition 3, $p<0.01$). Contrary to their mothers, they did not show cortisol responses when the other pair was manipulated and returned to the home cage (Condition 3A). They did, though, respond with significant cortisol elevations ($p<0.01$) when the other mother-infant pair was manipulated and removed from the cage (Condition 4A).

DISCUSSION

The overall levels of cortisol observed in this study are typical for the squirrel monkey, which is to say, roughly 10 times greater than in macaques or humans [5]. The high circulating levels of corticoids appear to result from a decreased binding affinity of cortisol receptors [5]. The half-life of cortisol in *Saimiri* has been estimated to be 79 min [4]. Thus, the 30-min sampling point used in the present study should be well suited for estimating significant changes in cortisol secretion that occurred between the time of initial manipulation and the time of blood sample collection.

Previous observations have indicated the ameliorating influence that contact between the primate mother and infant may have on the response of the pituitary-adrenal system [13]. The present study reveals some of the constraints under which such a phenomenon may occur. With separation and reunion of both pairs in the familiar home environment (Condition 1), no cortisol response was seen for either mothers or infants. Neither was there a significant increase in the normal activity level of the pairs, indicative of the low impact of the handling and separation procedure. However, when both pairs were reunited in the novel environment (Condition 2), infants showed significant increases in plasma cortisol concentrations. The contact with their mothers in this situation was not sufficient to produce an inhibition or buffering effect on the cortisol response. Though the cortisol levels of the mothers did not show a significant rise over basal levels in this condition, their activity levels were significantly depressed. This was mirrored by a marked increase in proximity-seeking behavior between the adult pairs, (i.e., Fig. 2). Proximity between the adults apparently had a functional consequence under these circumstances. Such an interpretation seems reasonable given that their endocrine responses did not differ from those in the Base condition, and would extend proximity-seeking from infants to adults as an effective coping response.

When a single dyad was manipulated and returned to the home cage, a marked increase in maternal activity was observed (Condition 3). There was an almost constant amount of movement and switching of positions from one side of the

cage to the other between the two pairs. Because of the apparent mutual nature of this behavior, this same agitation was reflected in the movement scores of the nonmanipulated mother (Condition 3A). In both cases their cortisol levels were also significantly elevated. However, this was not true for their infants who showed a significant cortisol elevation only when they were the manipulated pair, and not when in the nonmanipulated situation. This suggests that they were reacting to some different aspect in the situation, and that their responses were not simply a (transmitted) function of their mothers' increased state of agitation. They did, however, show a significant elevation as the nonmanipulated pair when the other dyad was removed to the novel environment (Condition 4A). Their mothers also responded with a similar 60% increase over basal levels in Condition 4A. When the single dyad was removed and reunited in the novel environment, both mother and infant showed significant cortisol elevations, while any movement on the part of the mothers virtually ceased.

Several interesting features emerge from above observations. First is the clear indication that behavioral and cortisol measures are not always concordant. Mothers may show a significant suppression of normal activity levels with no significant change in their cortisol (as in Condition 2). They may show a rise in activity level with an attendant rise in cortisol (Condition 3); or a decrease in activity and a rise in cortisol levels (Condition 4); or even no significant change in activity level and a significant elevation in cortisol (Condition 4A). These differences suggest that the elicited responses are situationally dependent and that the behavioral response may not necessarily reflect the internal "state" of the animal. In no case, however, was there a significant rise in activity level without an attendant rise in cortisol, which would seem to indicate that increased movement about the cage is reflecting an increased level of agitation on the part of the mother, although an absence of such overt behavioral responses may not indicate that there is no distress. The cortisol measure was the only indicator of the infant's response, since they maintained close physical contact with the mother in all conditions, never showing any overtly detectable signs of context-oriented "distress" responses.

We also obtained clear evidence for a lack of concordance between maternal and infant cortisol responses, despite the fact that mothers and infants remained in almost constant physical contact in all conditions. Mothers were not capable of buffering their infant's response in an unfamiliar situation. Thus, the infant's response is not merely reflective of the mother's state, and the two members of the dyad are apparently responding to different aspects of the situation. Vogt and Levine [15] have also reported a situation in which infants' responses did not track those of their mothers. Infants clearly showed elevated cortisol levels in response to disturbance by a stranger while in contact with their mothers, whereas no cortisol elevations were observed in their mothers. The infant response may have reflected disturbance or fear of the unfamiliar conspecific, but it may also reflect heightened arousal towards the novel social situation. The capacity for mothers and infants, and other social partners, to inhibit arousal under varying conditions is obviously complex and context oriented.

The predominant factor in eliciting responses from the mothers in this study appears to be social disruption. Significant behavioral and/or physiological responses occurred whenever their social partners were removed or one pair was manipulated (Conditions 3, 4, 3A and 4A). Indeed, social cues would seem to be a powerful modulator of arousal in these adults. In Condition 2, where movement was almost totally inhibited and the adults spent much of the time in close physical proximity, no significant elevation in cortisol was observed. The extent to which this behavior may provide an appropriate coping response and buffer endocrine changes over an extended period of time cannot, of course, be determined from this study. The results of Condition 4A indicate that separation from the other dyad elicited a significant cortisol elevation. The response does not appear to be due to disturbance, because there is greater disturbance in Condition 1 than in Condition 4A. Thus, the present study indicates that social separation, other than separation of a mother from her infant, is capable of eliciting a cortisol response in *Saimiri*.

In contrast to the mothers, infants appear to be particularly sensitive to novelty. This would seem a reasonable explanation of their elevated cortisol responses in Conditions 2 and 4. It is, however, more difficult to understand why they should show elevated responses in Condition 3, unless they were responding to the manipulation itself. This seems unlikely, as no elevated response was observed following manipulation in Condition 1. Perhaps the heightened agitation of their mothers, together with being the manipulated pair, was too potent a stressor for inhibition of the pituitary-adrenal system to occur. Finally, the significant cortisol response in Condition 4A indicates that the infants, like their mothers, were responding to removal of the other dyad. It may be that the close confinement imposed on the mother-infant pairs by the housing conditions of this study made removal of the other pair a more traumatic event for both mothers and infants than would otherwise have been the case.

TABLE 1

SUMMARY OF MOTHERS'S (M) AND INFANTS' (I) CORTISOL RESPONSES AND MOTHERS' MOVEMENT IN THE SEPARATION-REUNION CONDITIONS

Condition	Movement	Cortisol	
	M	M	I
1 Both Pairs—Home	ns	ns	ns
2 Both Pairs—Novel	↓	ns	↑
3 Single Pair—Home	↑	↑	↑
3A Nonmanipulated	↑	↑	ns
4 Single Pair—Novel	↓	↑	↑
4A Nonmanipulated	ns	↑	↑

ns—Change not significant.
↑↓—Significant change.

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